

Danmarks Tekniske Universitet

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Course title: Bayesian Machine Learning

Course number: 02477

Aids allowed: All aids except internet and AI/LLMs

Exam duration: 4 hours

Weighting: 100%

02477 Bayesian Machine Learning Exam 2026

Technical University of Denmark

- **Duration:** 4 hours
- **Aids:** All aids except internet and AI/LLMs
- **Results:** Report all numeric results with **2 digits** after the decimal point.
- **Explain** how you arrived at your results, document intermediate results when possible.
- **Hand-in:** Your solution can be either digitally as a PDF, on paper or both.
- **Student number:** Make sure your student number is **visible on all pages**.

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Part 1: Regression

Consider a regression model of the form:

$$y_n = f(x_n) + \epsilon_n, \quad (1)$$

where $x_n, y_n \in \mathbb{R}$ are inputs and outputs, respectively, $\epsilon_n \sim \mathcal{N}(0, \sigma^2)$ is i.i.d. Gaussian noise, and

$$f(x) = \theta_0 + \theta_1 \cos\left(2\pi\frac{1}{3}x\right) + \theta_2 \cos\left(2\pi\frac{2}{3}x\right) + \theta_3 \cos(2\pi x), \quad (2)$$

where $\boldsymbol{\theta} = [\theta_0 \ \theta_1 \ \theta_2 \ \theta_3]^T \in \mathbb{R}^4$ are the model parameters.

Assume the prior distribution for $\boldsymbol{\theta}$ is $\mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1}\mathbf{I})$, where $\mathbf{I}$ is the identity matrix and $\alpha > 0$ is the prior precision.

Below you are given a dataset of $N = 5$ observations:

$$\mathbf{x} = [-0.3 \quad 1.3 \quad -0.7 \quad 1.2 \quad -0.2], \quad (3)$$

$$\mathbf{y} = [2.1 \quad -1.0 \quad 0.7 \quad -0.7 \quad 1.7], \quad (4)$$

such that the n 'th entry in $\mathbf{x}$ is x_n and similar for $\mathbf{y}$.

Assume $\sigma = \frac{1}{10}$ and $\alpha = 1$.

Question 1.1: Determine the distribution $p(y_n|\boldsymbol{\theta}, x_n = 0)$.

Question 1.2: Compute the MAP estimate for $\boldsymbol{\theta}$ using the given dataset.

Question 1.3: Compute 95% intervals for the prior and exact posterior distributions of θ_1 .

Question 1.4: Determine the exact posterior predictive distribution $p(y^*|\mathbf{y}, x^*)$ for $x^* = 3$.

Question 1.5: Determine the analytical distribution $p(\mathbf{y})$ and evaluate $\log p(\mathbf{y})$ for the given dataset.

Part 2: Generalized linear models

A restaurant is analyzing their coffee sales, and an employee has collected a dataset consisting of the number of daily cups of coffee sold $y_n \in \{0, 1, 2, \dots\}$ as a function of the outside temperature $t_n \in \mathbb{R}$. Figure 1 shows a scatter plot of the dataset.

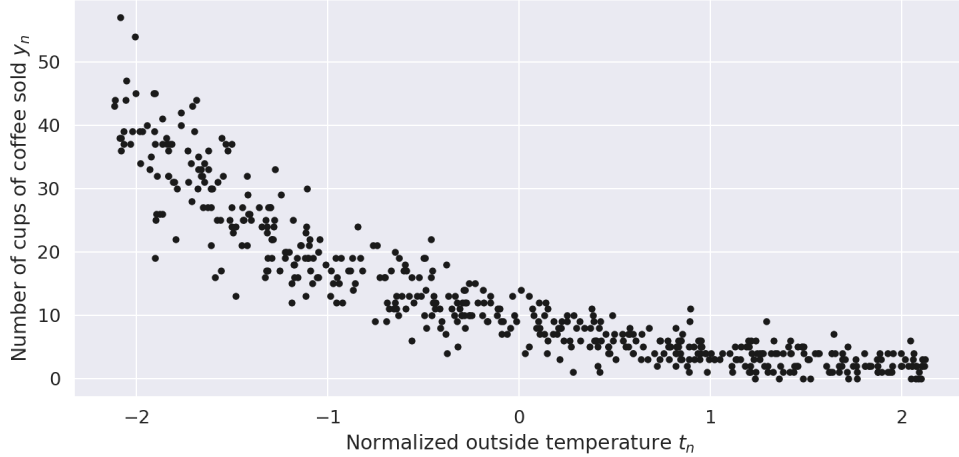


Figure 1: Scatter plot of number of cups of coffee sold as a function of the (normalized) outside temperature.

Consider the following model:

$$f(t_n) = w_0 + w_1 t_n, \quad (5)$$

where $\mathbf{w} = [w_0 \ w_1]^T \in \mathbb{R}^2$ are model parameters and $t_n \in \mathbb{R}$ is the normalized temperature for the n 'th observation.

Question 2.1: For the purpose of modelling the data in Fig. 1 using the model in eq. (5), choose an appropriate distribution for the likelihood and provide an appropriate link function if necessary. Justify your choices.

Assume a multivariate Gaussian prior distribution for $\mathbf{w}$:

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \mathbf{0}, \alpha^{-1} \mathbf{I}), \quad (6)$$

where $\alpha > 0$ is the prior precision and $\mathbf{I}$ is the identity matrix.

Question 2.2: Determine the prior precision such that the event $-3 \leq w_1 \leq 3$ has approximately 95% probability under the prior distribution.

Part 3: Markov Chain Monte Carlo

Let $p(\mathbf{x})$ be the probability density for a vector $\mathbf{x} = [x_1 \ x_2]^T \in \mathbb{R}^2$ given by

$$p(\mathbf{x}) \propto e^{-\frac{1}{2}\mathbf{x}^T \mathbf{x} - 20 \log(1 + e^{\mathbf{x}^T \mathbf{1}})}, \quad (7)$$

where $\mathbf{1} = [1 \ 1]^T$ is a vector of ones and

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\mathbf{x} - 20\sigma(\mathbf{x}^T \mathbf{1})\mathbf{1}, \quad (8)$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the logistic sigmoid function.

Consider the Metropolis algorithm with a Gaussian proposal distribution, $q(\mathbf{x}^* | \mathbf{x}^{k-1}) = \mathcal{N}(\mathbf{x}^* | \mathbf{x}^{k-1}, \frac{1}{2}\mathbf{I})$, where k is the iteration index for the Markov chain. Assume that the Markov chain is initialized at $\mathbf{x}_0 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ at iteration 0.

Question 3.1: Suppose the proposed candidate state is $\mathbf{x}^* = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$. Compute the probability of accepting $\mathbf{x}^*$ given the current state $\mathbf{x}_0$.

Question 3.2: Implement the Metropolis sampler for the distribution $p(\mathbf{x})$. Run a single chain initialized at $\mathbf{x}_0$. Run 1000 iterations and do not discard any warm up samples. Plot the trace.

Use the posterior samples of $\mathbf{x}$ from Question 3.2 to answer the next question. If you did not solve the previous question, you can draw 1000 samples from the following multivariate normal distribution instead:

$$\mathbf{x} \sim \mathcal{N}\left(\begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 & -\frac{1}{2} \\ -\frac{1}{2} & 2 \end{bmatrix}\right). \quad (9)$$

Question 3.3: Estimate $\mathbb{E}[x_1 x_2^2]$ using Monte Carlo estimation.

Consider now a Hamiltonian Monte Carlo (HMC) sampler for $p(\mathbf{x})$. Assume a standardized Gaussian distribution for the momentum variable, $\boldsymbol{\nu} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\boldsymbol{\nu}_0 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, and step-size $\eta = \frac{1}{2}$. Again, assume the chain is initialized at $\mathbf{x}_0 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ at iteration 0.

Question 3.4: Assuming the current state is $(\mathbf{x}_0, \boldsymbol{\nu}_0)$, compute the proposed state $(\mathbf{x}^*, \boldsymbol{\nu}^*)$ after a single leapfrog step, i.e. $L = 1$.

Now assume the proposed state is $(\mathbf{x}^*, \boldsymbol{\nu}^*)$ for $\mathbf{x}^* = [-1 \ -3]^T$ and $\boldsymbol{\nu}^* = [-1 \ -\frac{1}{2}]^T$.

Question 3.5: What is probability of accepting the state $(\mathbf{x}^*, \boldsymbol{\nu}^*)$ given the current state is $(\mathbf{x}_0, \boldsymbol{\nu}_0)$?

Part 4: Variational inference for a classification model

Consider the following model for binary classification:

$$y_i | \mathbf{w} \sim \text{Ber}(\sigma(w_0 + w_1 x_i)), \quad (10)$$

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (11)$$

where $x_i \in \mathbb{R}$ is an input feature, $y_i \in \{0, 1\}$ is the target, and $\sigma(x) = \frac{1}{1+e^{-x}}$ is the logistic sigmoid function.

Below you are given two different point estimates for some dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ (data not given here):

$$\hat{\mathbf{w}}_A = \begin{bmatrix} 2.09 \\ -0.89 \end{bmatrix}, \quad \hat{\mathbf{w}}_B = \begin{bmatrix} 2.42 \\ -1.01 \end{bmatrix}. \quad (12)$$

One of the two estimates is the maximum likelihood estimate and one is the maximum a posteriori (MAP) estimate for the model above.

Question 4.1: Which of the two point estimates above is the MAP estimate? Justify your answer.

Question 4.2: Compute the posterior predictive distribution $p(y^* | \mathbf{y}, x^*)$ for $x^* = 1$ using the plug-in approximation based on the $\hat{\mathbf{w}}_A$ point estimate.

Consider the variational family $\mathcal{Q}$ of distributions given by:

$$q(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \mathbf{m}, \tau^2 \mathbf{I}), \quad (13)$$

where $\mathbf{m} \in \mathbb{R}^2$ is the variational parameters and $\tau^2 = \frac{1}{10}$ is a fixed constant.

Question 4.3: Compute the posterior predictive distribution $p(y^* | \mathbf{y}, x^*)$ for $x^* = 1$ using the variational approximation $q(\mathbf{w})$ for $\mathbf{m} = \begin{bmatrix} 2 & -1 \end{bmatrix}^T$ and Monte Carlo estimation.

Let $p(\mathbf{w} | \mathbf{y})$ denote the posterior distribution and let $q^*(\mathbf{w}) = \mathcal{N}(\mathbf{w} | \mathbf{m}^*, \tau^2 \mathbf{I})$ denote the optimal variational approximation such that

$$\mathbf{m}^* = \arg \min_{q \in \mathcal{Q}} \text{KL}[q(\mathbf{w}) || p(\mathbf{w} | \mathbf{y})] = \arg \min_{\mathbf{m} \in \mathbb{R}^2} \text{KL}[q(\mathbf{w}) || p(\mathbf{w} | \mathbf{y})] \quad (14)$$

Question 4.4: Determine the entropy of the optimal variational approximation q^* .

Part 5: Gaussian processes

Consider the following regression model for $y_n \in \mathbb{R}$ and $\mathbf{x}_n \in \mathbb{R}^2$:

$$y_n = f(\mathbf{x}_n) + \epsilon_n, \quad (15)$$

where $\epsilon_n \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \frac{1}{3}$.

Assume a Gaussian process prior such that $f \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$ for $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^2$ with the following covariance function

$$k(\mathbf{x}, \mathbf{x}') = 4 \exp(-\|\mathbf{x} - \mathbf{x}'\|^2). \quad (16)$$

Below you are given a training data matrix $\mathbf{X}$ containing $N = 4$ input vectors:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \mathbf{x}_3^T \\ \mathbf{x}_4^T \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad (17)$$

with corresponding targets:

$$\mathbf{y} = [-0.25 \quad 0.42 \quad 0.49 \quad -1.17]^T \quad (18)$$

such that the second observation has input vector $\mathbf{x}_2 = [0 \quad 1]^T$ and target $y_2 = 0.42$.

Let $y^* = f(\mathbf{x}^*) + \epsilon^*$ for $\mathbf{x}^* = [\frac{1}{2} \quad \frac{1}{2}]^T$ and $\epsilon^* \sim \mathcal{N}(0, \sigma^2)$.

Question 5.1: Determine the prior predictive distribution $p(y^*|\mathbf{x}^*)$.

Question 5.2: Determine the posterior predictive distribution $p(y^*|\mathbf{y}, \mathbf{x}^*)$.

Let $\mathbf{f} = [f(\mathbf{x}_1) \quad f(\mathbf{x}_2) \quad \dots \quad f(\mathbf{x}_N)]$ and suppose $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$.

Question 5.3: Determine the distribution $p(\mathbf{f}^*|\mathbf{f}, \mathbf{x}^*)$ for $\mathbf{f}^* = f(\mathbf{x}^*)$.

Finally, let $\mathbf{f}_2$ be a transformation of $\mathbf{f}$ such that

$$\mathbf{f}_2 = \mathbf{A}\mathbf{f} + \mathbf{c}, \quad (19)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is a constant matrix and $\mathbf{c} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

Question 5.4: Determine the distribution $p(\mathbf{f}_2)$.